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# Probability and Statistics (MT-2005)

## Project Report

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**Submitted to:**

Ms. Rijah Khan

**Submitted by:**

Muhammad Abdullah (22i-2672)

Muhammad Ismail (22i-2483)

BS-SE(E)





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## 1.Introducton

The purpose of this document is to report on the research correlating precipitation with other environmental factors. The project aims to utilize the potential of **R-language and Studio** in data analysis. It is another exuberant example of how probability and statistical analysis can help solve real world problems.

For our research we have used a dataset from Kaggle (Allushi, n.d.). The dataset showcases sufficive weather data from the coastal city of Vlore, Albania. It holds **average temperature, minimum temperature, maximum temperature, wind direction, wind speed, millimeters of precipitation, and the pascals of pressure experienced by the city throughout 2021**. Our regression model is strictly a preliminary research artefact, defined in the aim to aid a greater understanding of determining the precise measurement of precipitation and the precise date at which a city or a major metropolitan area would experience that precipitation.

## 2.Literature Review

The reservation of water has become an essential area of concern, now more than ever before. With the rising global temperature and climate change, the word's water only becomes a more vital and a greater pivot. It is reported that water scarcity effects 1/5<sup>th</sup> of the world's population (www.un.org, 2015).

Only 3% of natural water resources is drinking water, of which 20% is trapped in glaciers which are melting, leading to flooding, putting a greater stress on available water resources. In the midst of a water crisis, rainwater can as a reliable water reserve. Many people have already started using rainwater for menial tasks such as for flushing and washing purposes. However, there's great potential in the utilization of rainwater, and this regression model is certainly a greater step in a revolutionary direction. At a domestic level, rainwater harvesting allows households to become less reliant on municipal water supplies and can provide a sustainable secondary water source for gardening, washing cars, flushing toilets,





and even laundry with proper treatment. Moreover, it reduces water bills and the strain on shared water resources.

Commercially, rainwater harvesting can be a game-changer. Industries such as agriculture, which require significant amounts of water, can substantially reduce their operational costs and reliance on external water supplies. Moreover, properties like hotels and office buildings can use harvested rainwater for landscaping and sanitation purposes, promoting sustainability while reducing overheads. (Rainwater as a solution to water scarcity: case studies from around the world, 2024)

### 3. Regression Model and Variables

In our research for the project we applied many regression models to explore the relationships between the contents of the weather of Vlore. In our model, we kept the Precipitation experienced each day in millimeters as independent, other factors such as maximum temperature, wind speed, atmospheric pressure, wind direction and minimum.

#### 3.1 Description

**Precipitation (mm)** : the millimeters of rainfall experienced every day.

**Maximum Temperature (\*C)**: the highest recorded temperature of the day.

**Minimum Temperature (\*C)**: the lowest recorded temperature of the day.

**Wind Speed (Km/h)**: how fast the wind is blowing every day.

**Wind Direction**: the direction in which the wind is blowing

**Pressure (Pascals)** : the atmospheric pressure on a given day.

All data is accurate to the city of Vlore, Albania circa 2021.





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We quantify that, that these variables have a relation with precipitation experienced by the city.

## 4.Description, Results and Discussion

### 4.1.General overview

```
Rows: 365 Columns: 7
— Column specification —
Delimiter: ","
chr (1): date
dbl (6): tmin, tmax, prcp, wdir, wspd, pres
```

The names of the variables are shortened.

### 4.2. Task2

Following is the summary of the data,

```
> summary(vlore_data_1)
  date      tmin      tmax      prcp      wdir      wspd
Length:365  Min.   :-3.40  Min.    : 2.70  Min.    : 0.000  Min.    :  1.0  Min.    : 3.800
Class :character  1st Qu.: 7.10  1st Qu.:16.60  1st Qu.: 0.000  1st Qu.: 44.0  1st Qu.: 6.200
Mode  :character  Median :12.00  Median :21.60  Median : 0.000  Median : 93.0  Median : 7.500
      Mean  :12.27  Mean  :22.36  Mean  : 3.035  Mean  :116.9  Mean  : 8.266
      3rd Qu.:17.55  3rd Qu.:28.25  3rd Qu.: 1.400  3rd Qu.:164.0  3rd Qu.: 9.400
      Max.   :25.90  Max.   :39.40  Max.   :48.000  Max.   :359.0  Max.   :21.200
      NA's    :2     NA's    :2     NA's    :2     NA's    :2     NA's    :2

  pres
Min.   : 999.7
1st Qu.:1011.6
Median :1014.7
Mean   :1015.2
3rd Qu.:1018.6
Max.   :1032.5
NA's   :2
```

For more accurate and purposeful data set, we excluded days when Vlore experience less than 1 mm of precipitation.

```
> vlore_data_filtered <- vlore_data_1 %>%
```





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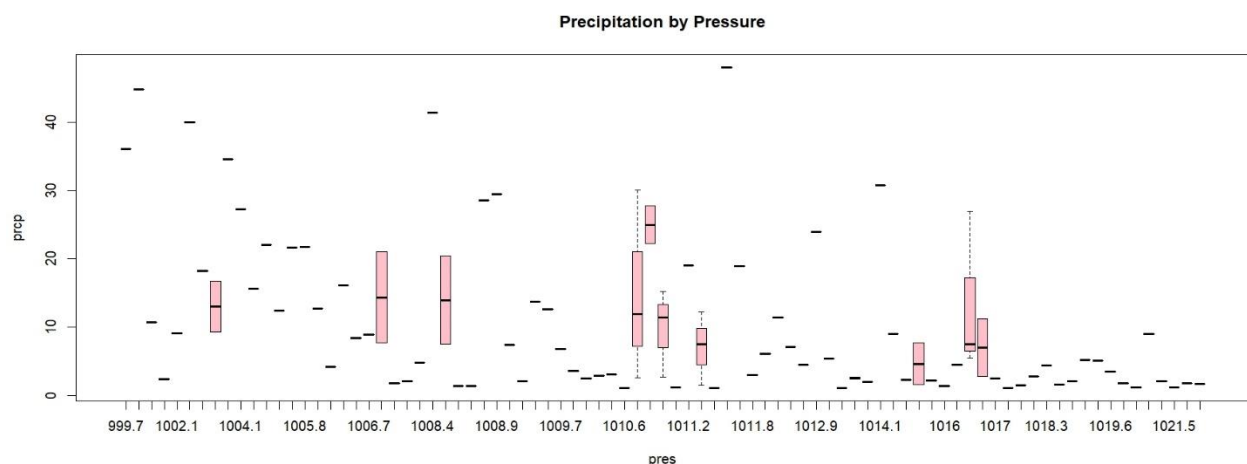
```
+ filter(prcp >= 1)
```

the modes of the filtered data:

```
Mode of Precipitation: 1.1
> cat("Mode of Max Temperature:", tmax_mode, "\n")
Mode of Max Temperature: 17 16.6
> cat("Mode of Min Temperature:", tmin_mode, "\n")
Mode of Min Temperature: 7.2
> cat("Mode of Wind Direction:", wdir_mode, "\n")
Mode of Wind Direction: 32 31
> cat("Mode of Wind Speed:", wspd_mode, "\n")
Mode of Wind Speed: 6
> cat("Mode of Atmospheric Pressure:", pres_mode, "\n")
Mode of Atmospheric Pressure: 1012.9
```

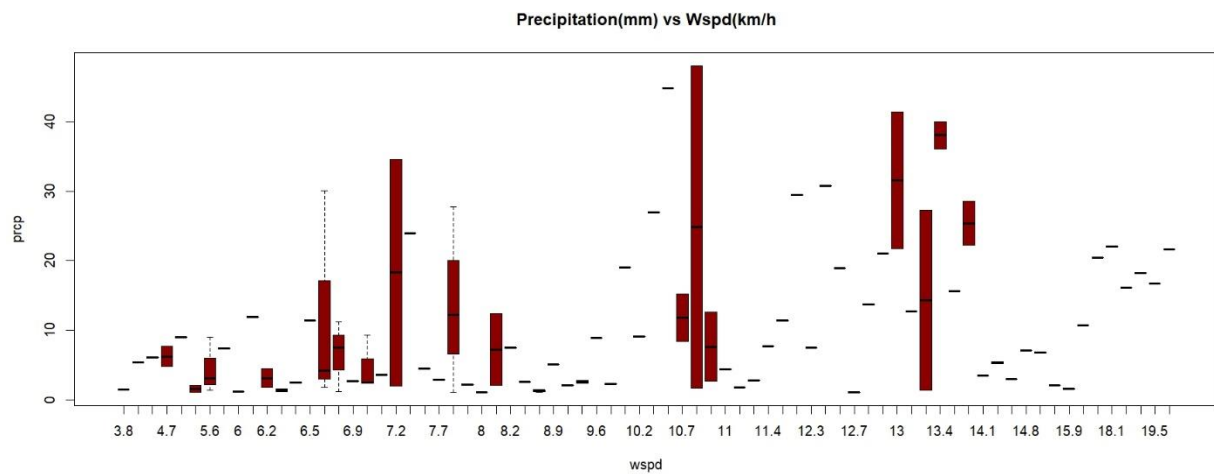
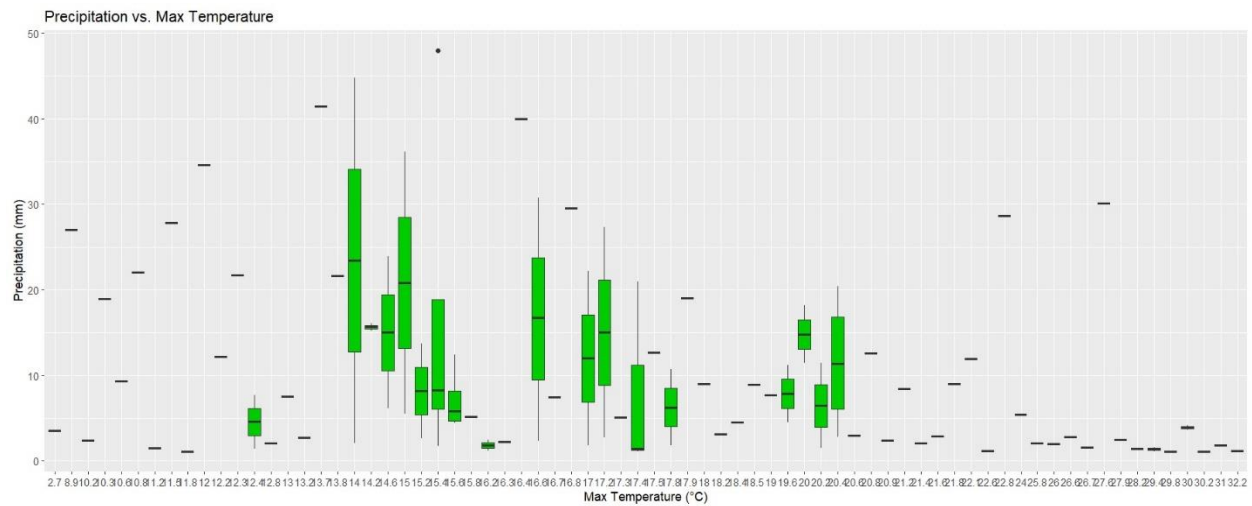
- The mode values of all involved datasets, point towards Vlore's coastal climate, however **the 16.6 \*C maximum temperature** suggests the city is chillier than most and **7.2 \*C minimum temperature mode** confirms modernity.
- Clearly, not as windy with only 6, but it is **higher than most cities**.
- Normal Precipitation mode, but the data has clear outliers and we have studied these

### 4.3. Task 3





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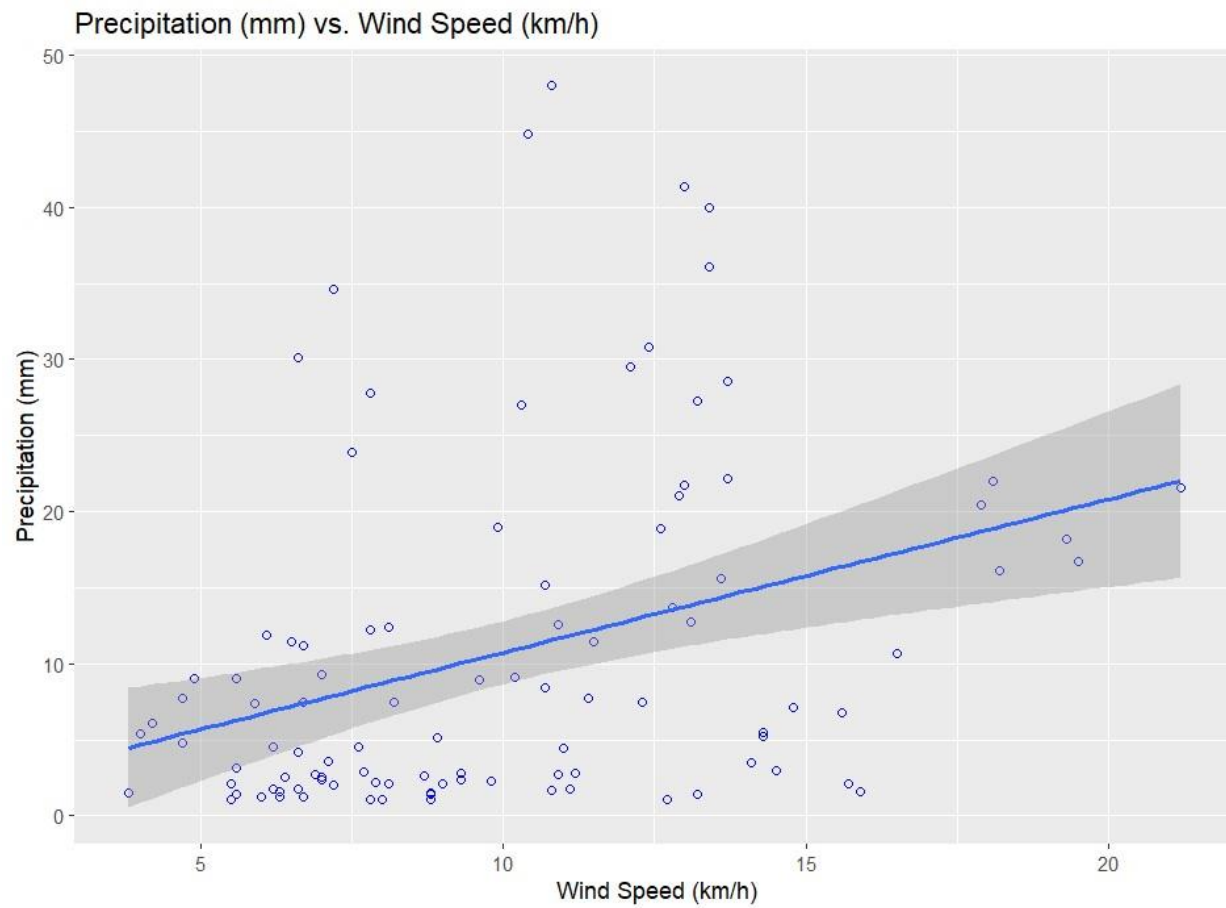
The longest whisker suggest the wind speeds, pressures and temperature for which vlore experince th elarest range of precipitation. The outliers are the values for which there is precipitation, but outside of the IQR. The single plots are for the values, when there usually isn't any precipitation.







## 4.4 Task 4

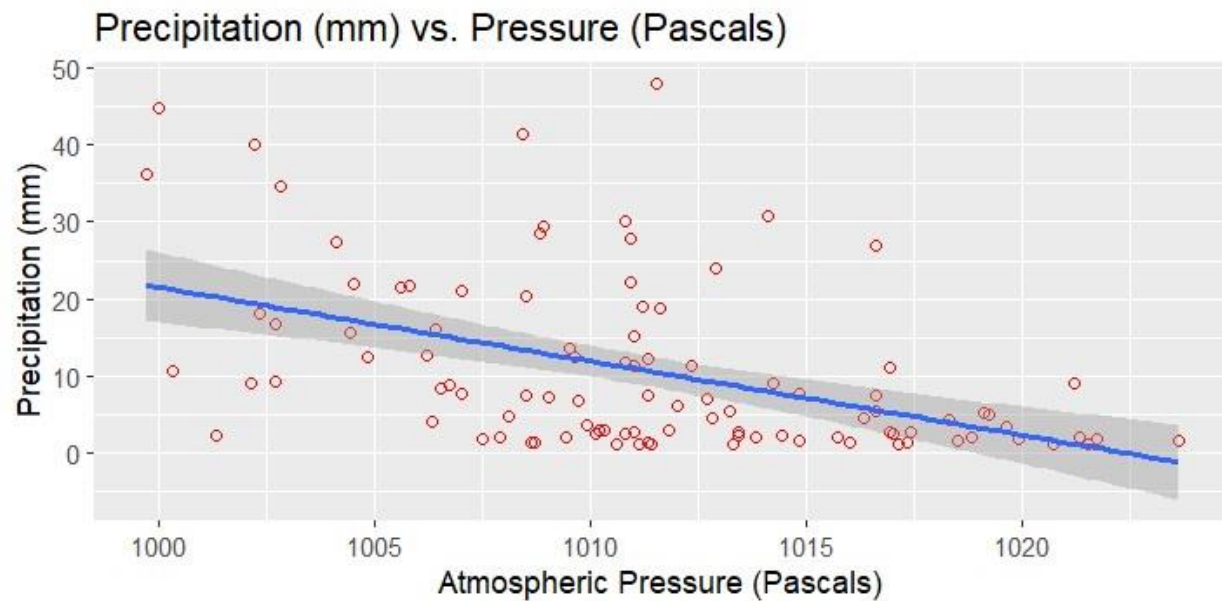


Weaker correlation, but in general  
a windy rainy day experiences more precipitation than the still rainy day





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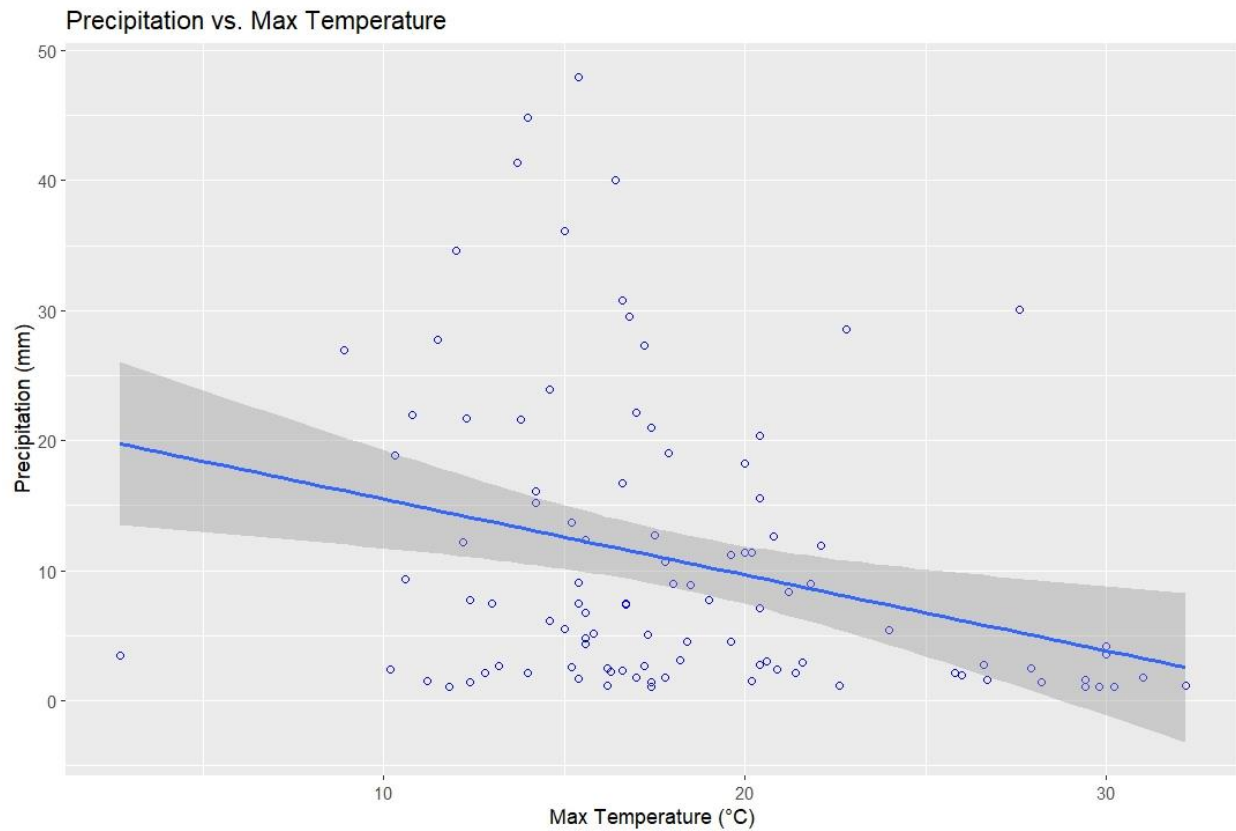
Coefficient of determination: 0.15

Suggest a weaker correlation, because precipitation is largely affected by other factors, not available in the dataset. In general, we can say the rainfall, when atmospheric pressure is higher, will yield less precipitation.





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Coefficient of Variance : 0.167

Negative correlation, and this is also weak, because factors unrecorded have a greater influence on the precipitation e.g(frequency of precipitation, and the amount of water vapor in the air relevant to windspeed). In general, on rainy days with higher max temperature the precipitation is lower, than on rainy days with a lower max temperature.





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RM with two independent variables:

```
lm(formula = prcp ~ tmax + wspd, data = vlore_data_1)
```

Residuals:

| Min     | 1Q     | Median | 3Q    | Max    |
|---------|--------|--------|-------|--------|
| -13.159 | -3.181 | -0.988 | 0.939 | 41.190 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 0.01689  | 1.69411    | 0.010   | 0.992        |
| tmax        | -0.20630 | 0.04791    | -4.306  | 2.14e-05 *** |
| wspd        | 0.92316  | 0.12310    | 7.499   | 5.05e-13 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.738 on 360 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.2139, Adjusted R-squared: 0.2095

F-statistic: 48.98 on 2 and 360 DF, p-value: < 2.2e-16

RM with more than two independent variables:

Residuals:

| Min     | 1Q     | Median | 3Q    | Max    |
|---------|--------|--------|-------|--------|
| -13.425 | -3.074 | -0.524 | 1.600 | 36.553 |

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | 454.751737 | 68.602986  | 6.629   | 1.25e-10 *** |
| tmax        | -0.972169  | 0.136427   | -7.126  | 5.75e-12 *** |
| wspd        | 0.452174   | 0.119793   | 3.775   | 0.000188 *** |
| wdir        | -0.003359  | 0.003621   | -0.928  | 0.354134     |
| pres        | -0.436746  | 0.067288   | -6.491  | 2.86e-10 *** |
| tmin        | 0.820987   | 0.164512   | 4.990   | 9.43e-07 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.861 on 357 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.4102, Adjusted R-squared: 0.4019

F-statistic: 49.66 on 5 and 357 DF, p-value: < 2.2e-16





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In this case wind direction is an insignificant variable

## 5. References

### Works Cited

(2015). Retrieved from [www.un.org](https://www.un.org/waterforlifedecade/scarcity.shtml): <https://www.un.org/waterforlifedecade/scarcity.shtml>

Allushi, M. (n.d.). *Weather of Albanian cities in the last 3 years*. Retrieved from Kaggle: <https://www.kaggle.com/datasets/gucci1337/weather-of-albania-last-three-years/code>

*Rainwater as a solution to water scarcity: case studies from around the world*. (2024, May 11). Retrieved from [smartwateronline.com](https://smartwateronline.com/news/rainwater-as-a-solution-to-water-scarcity-case-studies-from-around-the-world#:~:text=Utilising%20rainwater%20tanks%2C%20homeowners%20can,down%20on%20household%20water%20bills): <https://smartwateronline.com/news/rainwater-as-a-solution-to-water-scarcity-case-studies-from-around-the-world#:~:text=Utilising%20rainwater%20tanks%2C%20homeowners%20can,down%20on%20household%20water%20bills>.

